

Experiment No . 02

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Batch: T22

Roll no: 101

Aim: To understand version control system/ source code management, install git and create a github account

Theory:

Version Control System (VCS) / Source Code Management (SCM)

A **Version Control System** (VCS) is a tool that helps developers manage changes to their codebase over time. It tracks revisions of files, allows developers to collaborate on code without overwriting each other's work, and provides a history of changes. This makes it easier to revert to earlier versions, compare code differences, and maintain an organized workflow.

There are two main types of VCS:

1. **Centralized VCS:** The code is stored in a central repository. Developers check out and check in changes to this single location.
2. **Distributed VCS:** Every developer has a full copy of the code repository, allowing for greater flexibility and offline work. Git, a distributed VCS, is the most popular system used today.

Git

Git is a distributed version control system that allows multiple developers to work on a project simultaneously. It tracks changes to files, creates snapshots of the entire repository at different points in time, and supports branching and merging, making it ideal for collaborative work.

Basic Git Concepts:

- **Repository:** A project's directory where all code and version history are stored.
- **Commit:** A snapshot of changes made to the repository.
- **Branch:** A separate line of development where changes can be made without affecting the main project.
- **Merge:** Combining changes from different branches back into a main branch.
- **Clone:** A copy of the repository, typically from a remote source like GitHub.
- **Push/Pull:** Sending and receiving updates between local repositories and

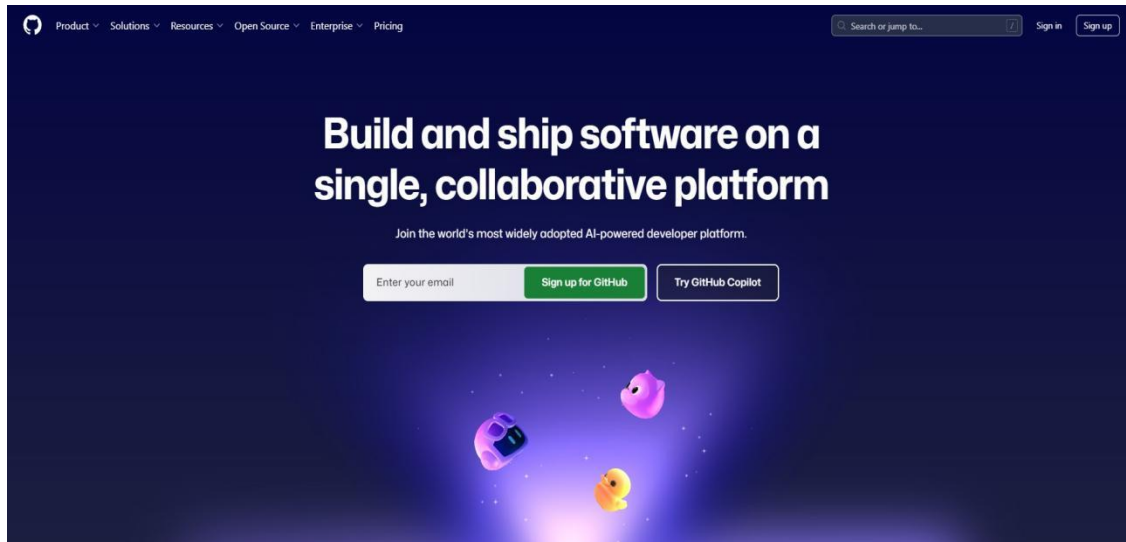
remote repositories.

GitHub

GitHub is a cloud-based hosting service for Git repositories. It allows developers to store their code online, collaborate with others, and manage issues, pull requests, and documentation. GitHub also provides an interface to interact with Git repositories, making it easier for developers to contribute to projects.

Steps to create GitHub account:

1. Go to github.com and click **Sign Up**.



2. Enter your **username**, **email**, and **password**.

Create your free account

Explore GitHub's core features for individuals and organizations.

[See what's included](#)

Already have an account? [Sign in](#)

Sign up to GitHub

Email*

Password*

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your country*

India

For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

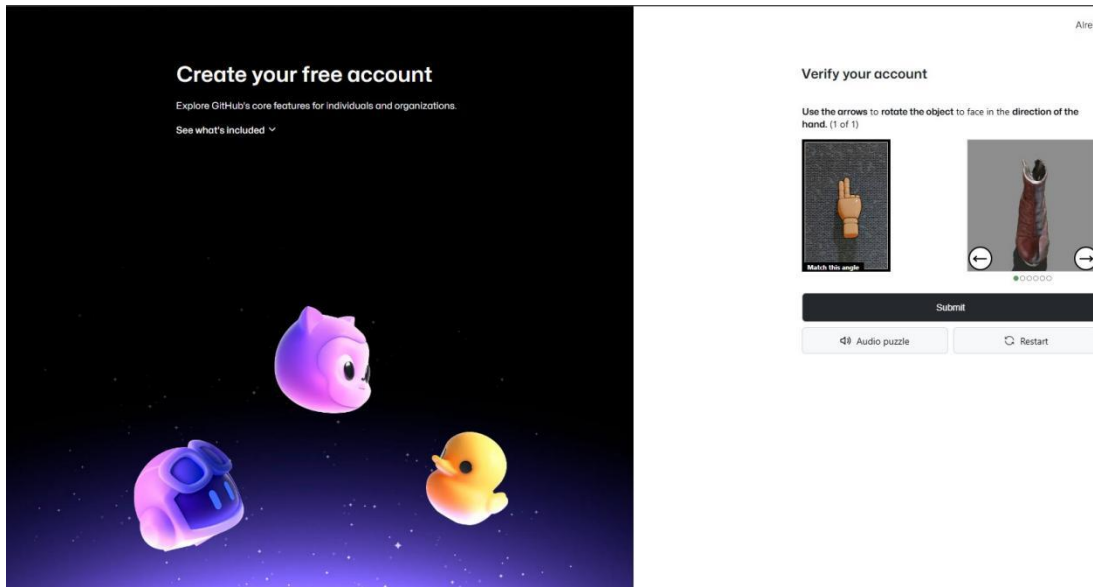
Email preferences

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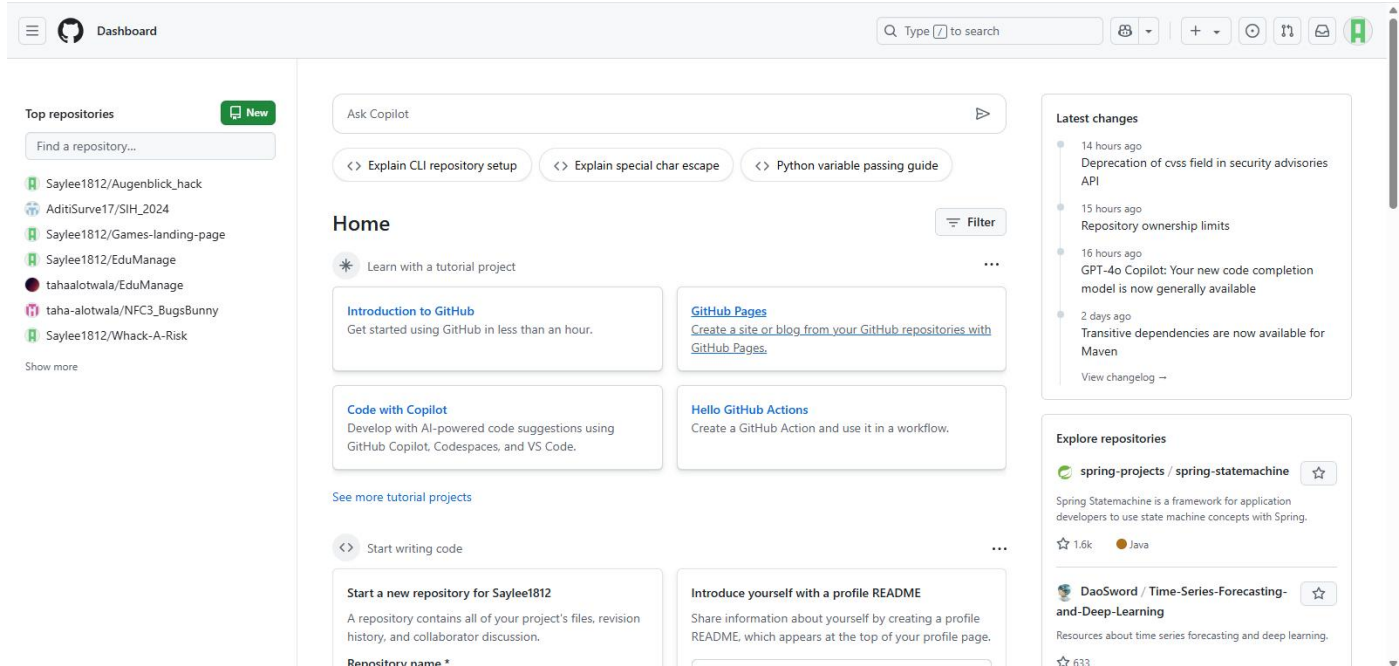
By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

3. Complete the verification puzzle.

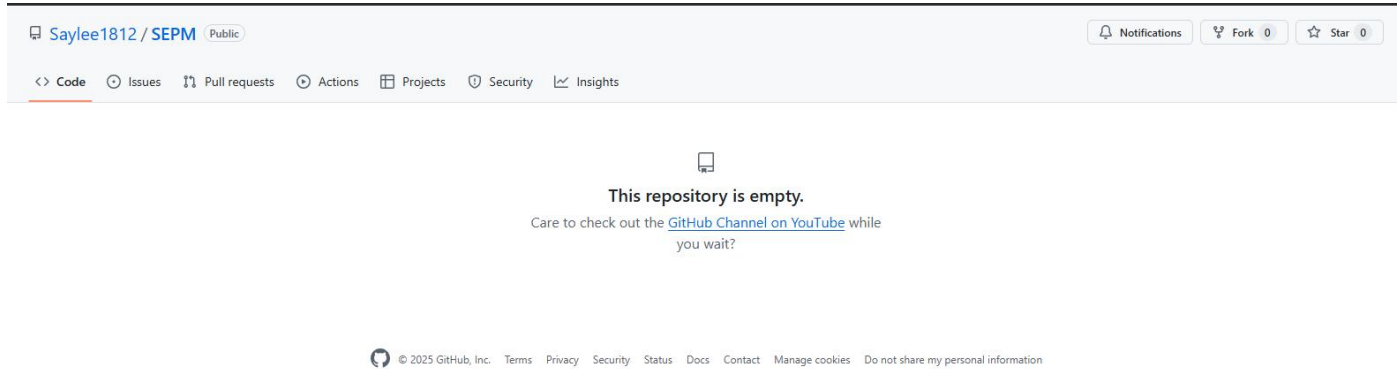


4. Verify your email through the link sent to your inbox.

5. Log in to your new GitHub account.



6. Create SEPM repository..



Conclusion: Hence we have successfully learnt version control system/ source code management, install git and create a github account